

Feiran Wang

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RESEARCH FOCUS & INTERESTS

- **Robotics:** Vision-Language-Action Model, World Action Model, Long-Horizon Manipulation
- **Spatial Intelligence:** Spatial Reasoning, Dynamic 3D Reconstruction, Self-Improving Agents
- **World Models:** Video Diffusion Model, Controllable Generation, Synthetic Data Generation

EDUCATION

- **University of Illinois Chicago** Aug. 2026 – Current
Ph.D. Student in Computer Science, Advisor: Prof. Yan Yan Chicago, United States
Prior doctoral study at Illinois Institute of Technology Jan. 2024 – Jul. 2026
- **University of Michigan Ann Arbor** Jan. 2026 – May 2026
Visiting Ph.D. Student, Advisor: Prof. Dawen Cai Ann Arbor, United States
- **University of Illinois Urbana-Champaign** Aug. 2022 – Dec. 2023
M.Eng. in Robotics, Advisor: Prof. David Forsyth Urbana, United States
- **Shanghai University** Aug. 2018 – Jul. 2022
B.S. in Computer Science, Advisor: Prof. Xiaoqiang Li Shanghai, China

PUBLICATIONS

*=Equal contribution | ‡=Author mentored by me

Robotics & Navigation

- [1] Feiran Wang, Zezhou Shang, Gaowen Liu, Yan Yan. **RayMap3R: Inference-Time RayMap for Dynamic 3D Reconstruction**. ECCV 2026.
- [2] Feiran Wang, Junyi Wu, Dawen Cai, Yuan Hong, Yan Yan. **CogniMap3D: Cognitive 3D Mapping and Rapid Retrieval**. ICLR 2026.
- [3] Zhe Tao[‡], Feiran Wang, Gaowen Liu, Ramana Rao Kompella, Yan Yan. **ForeAct3D: Policy-Grounded Future World Modeling for Vision-Language-Action Policies**. Under Review, 2026.
- [4] Wenxin Rong[‡], Haoxuan Wang, Feiran Wang, Yasutomo Kawanishi, Yan Yan. **Seeing Farther to Act Now: Long-Horizon Future Modeling for Robot Manipulation**. Under Review, 2026.
- [5] Junyi Wu, Van Nguyen Nguyen, Benjamin Planche, Jiachen Tao, ..., Feiran Wang, Terrence Chen, Yan Yan, Ziyang Wu. **Consistent Instance Field for Dynamic Scene Understanding**. CVPR 2026.
- [6] Jing Gao[‡], Wei Wang, Feiran Wang, Yan Yan. **LIST3R: Long-sequence Instance-aware 3D Reconstruction**. Under Review, 2026.

Spatial & Self-Evolving Agents

- [7] Feiran Wang, Xiaoqi Wang, Ziwei Li, Wenbin He, Yan Yan, Liu Ren. **From Sight to Foresight: Predictive Spatial Reasoning in Vision-Language Models**. Under Review, 2026. *Bosch internship project*
- [8] Vickie Chen[‡], Feiran Wang, Yasutomo Kawanishi, Yan Yan. **SEVA: Self-Evolving Vision-Language-Model Agent for Video Segmentation**. Under Review, 2026.

3D Vision & World Modeling

- [9] Feiran Wang*, Jiachen Tao*, Junyi Wu*, Haoxuan Wang, Bin Duan, Zongxin Yang, Yan Yan. **X-Field: A Physically Informed Representation for 3D X-ray Reconstruction**. NeurIPS 2025 (★Spotlight★).
- [10] Feiran Wang, Bin Duan, Jiachen Tao, Nikhil Sharma, Gaowen Liu, Dawen Cai, Yan Yan. **ZECO: ZeroFusion Guided 3D MRI Conditional Generation**. MVA 2025 (Oral).
- [11] Feiran Wang, Bin Duan, Junyi Wu, Gaowen Liu, Yan Yan. **Kepler4D: Controllable Future Video Generation via 4D Scene State Evolution**. Under Review, 2026.
- [12] Jiachen Tao, Benjamin Planche, ..., Feiran Wang, ..., Terrence Chen, Yan Yan, Ziyang Wu. **From Particles to Fields: Reframing Photon Mapping with Continuous Gaussian Photon Fields**. Under Review, 2025.
- [13] Feiran Wang, Xiaoqiang Li, Jitao Liu. **PCCN-RE: Point Cloud Colourisation Network Based on Relevance Embedding**. IET Computer Vision, 2022.

EXPERIENCE

- **Research Scientist Intern** May 2026 – Aug. 2026
Bosch Center for Artificial Intelligence, Sunnyvale, CA, Intern
 - **Predictive Spatial Reasoning:** Developed SpatialMind, a VLM combining metric depth adaptation, geometry-motion fusion, and progressive state-chain reasoning to advance spatial intelligence.
 - **Data Engine & Benchmark:** Built a geometry-grounded data engine for driving and egocentric videos, generating **30,677 training QA pairs** and a **2,000-question benchmark** covering eight spatial reasoning tasks.
 - **Model Performance:** Achieved a **52.65%** overall score on SpatialMind-2K, **+6.95 percentage points** over the same-data Qwen3-VL-8B SFT baseline, with competitive zero-shot results on VSI-Bench, OSI-Bench, and VLM4D.
- **Robotics Research & Development Engineer** May 2023 – Aug. 2023
Foxconn Interconnect Technology, San Jose, CA, Intern
 - **Simulation Development Platform:** Built Blender-to-MuJoCo toolchain for automatic URDF generation with joint dynamics calibration and collision geometry definition for rapid prototyping.
 - **Motion Planning & Control:** Integrated MoveIt for trajectory planning and collision detection in MuJoCo; designed PD controllers with real-time state feedback for precise trajectory tracking; developed visualization dashboard for monitoring joint positions, velocities, and torques.
 - **Sim-to-Real Deployment:** Optimized control parameters through simulation-hardware closed-loop iteration; validated grasping tasks on low-cost 3D-printed hardware platform.
- **Algorithm Engineer** Apr. 2022 – Aug. 2022
NPLACE Startup, Shanghai, China, Intern
 - **3D Reconstruction:** Independently explored pure vision reconstruction as alternative to company's LiDAR solution; built end-to-end pipeline through MVS reconstruction to point cloud completion.
 - **MVS Network Optimization:** Applied network pruning and multi-scale feature aggregation to MVSNet; compressed model via knowledge distillation achieving 10% inference speedup.

PROJECTS

- **Autonomous Vehicle Perception System** Sep. 2022 – Dec. 2022
UIUC, Advised by Prof. David Forsyth [Project Video](#)
 - **End-to-End Autonomous Driving System:** Built complete ROS-based autonomous driving system on real vehicle platform covering perception, planning, and control.
 - **Multimodal Perception & Localization:** Integrated YOLO detection for obstacle recognition and emergency braking; combined learning with Kalman filtering for lane tracking; implemented LiDAR SLAM.
- **3D Medical Image Analysis** Sep. 2024 – May 2026
University of Michigan Ann Arbor, Prof. Dawen Cai [Project Page](#)
 - **3D Medical Data Generation:** Designed ControlNet-conditioned diffusion model for controllable 3D volume generation, achieving **50×** synthetic data expansion to address medical image annotation scarcity.
 - **Semi-Automatic Annotation Platform:** Built interactive 3D annotation tool based on nnInteractive and nnUNet with morphological processing and connected component analysis, improving annotation efficiency.
 - **Large-Scale Distributed Training:** Deployed **240-GPU** multi-node training environment on HPC cluster with MPI optimization for large-batch parallel processing.

SKILLS

- **Robotics & Simulation:** ROS, MuJoCo, MoveIt, URDF, Blender, Motion Planning, PID Control
- **Programming Languages:** Python, C++, C#, MATLAB, JavaScript
- **Deep Learning & Training:** PyTorch, CUDA, TensorFlow, HPC Distributed Training
- **3D Vision & Reconstruction:** Open3D, OpenCV, SLAM, NeRF, 3DGS, VR

ACADEMIC SERVICES & ACTIVITIES

Conference Reviewer: ICLR 2026, NeurIPS 2026, ECCV 2026, CVPR 2026, ICMR 2026, ICCV 2025, MVA 2025
Journal Reviewer: Computer Vision and Image Understanding, Computers & Graphics
Invited Talk: Midwest Machine Learning Symposium (MMLS), 2025: Medical Imaging 3D Reconstruction

HONORS AND AWARDS

Cyrus Tang Scholarship, Illinois Institute of Technology	2024 – 2025
International Exchange Scholarship, Shanghai University	2020 – 2021
Academic Excellence Scholarship, Shanghai University	2019 – 2020